

[sigconf]acmart
booktabs graphicx subcaption
2024 2024 acmcopyright [CSS 586]Deep Learning ProjectsSpring 2024Bothell, WA, USA Deep Learning
Projects, Spring 2024, Bothell, WA, USA 15.00 10.1145/nnnnnnnn.nnnnnnnn 978-x-xxxx-xxxx-x/YY/MM
document Optimizing Zero-Shot Model Compression Pipelines
GitHub Copilot University of Washington Bothell 18115 Campus Way NE Bothell WA 98011 copilot@uw.edu

abstract Modern deep learning models, while powerful, are often too large and slow for deployment on resource-constrained edge devices. Model compression techniques like quantization, pruning, and knowledge distillation offer a solution by reducing model size and latency. However, the optimal combination and ordering of these techniques are not well understood, and most methods require extensive retraining on large datasets. This paper explores the landscape of zero-shot model compression, where the goal is to optimize models with little to no training data. We introduce a flexible ‘CompressionPipeline’ orchestrator to systematically evaluate various compression strategies. Through experiments on ResNet-18 for image classification (CIFAR-10) and BERT for natural language processing (GLUE MRPC), we analyze the trade-offs between model size, latency, and accuracy. Our findings demonstrate that multi-stage pipelines, such as combining pruning with quantization, can achieve significant size reductions (up to 75%) while maintaining near-original accuracy, all without requiring fine-tuning. We also present a grid search methodology for automatically discovering effective pipelines and profile performance on both CPU and GPU, providing a comprehensive guide for practitioners seeking to deploy efficient models in data-scarce environments.